Human Natural Structure

Social Meta-Structure Specification

The Six-Layer Grounding Stack from Body to Civilization

Volume 4 · Human Natural Structure Series

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Natural Structure Works (NSW)

A Note on Method and Evidence

This volume specifies the grounding stack of Human Natural Structure: the six Social Meta Layers that constitute Axis 3, the axis on which a generated output is checked against the structure of the human world, from the body to the civilization in which it acts. As in the preceding volumes, its claims fall into three registers, and the reader is entitled to know which is which.

Established. The traditions the stack draws on are real and well developed: embodied cognition and the symbol-grounding problem, the theory of communicative action, institutional theory, general systems theory and cybernetics, predictive coding and the free-energy principle, and perceptual control theory. Where this volume reports their content it reports established material the reader can verify in the sources cited.

HNS design. The arrangement of these traditions into a six-layer grounding stack, the composite grounding function, and the use of the stack as an external grounding check that feeds the control loop of the analytical engine are engineering decisions of HNS. They are internally well defined and, I argue, useful; they are not the only possible decomposition of social grounding.

Conjecture. That checking outputs against these six layers measurably improves the social alignment of a deployed system is a hypothesis, motivated and partially supported but not established here, and marked as conjecture where it appears.

A candid limitation, stated up front

This volume owes the reader an unusually direct caveat, because the stack it specifies is the least settled part of the framework. The six Social Meta Layers are a useful scaffold, not a proven taxonomy. Three weaknesses are acknowledged openly and revisited in the final chapter. First, the stack mixes two organizing principles: a scale (from the somatic body up to the civilization) and a set of domains (communicative, organizational, economic, normative). These do not cleanly coincide, and the ordering should be read as broadly scale-like rather than strictly monotone. Second, the layers are not mutually exclusive: the economic and the governance layers overlap, as do the protocol and the organizational, and the failure they suppress is in one case shared. Third, the number six is a chosen resolution that keeps this axis aligned with the rest of the framework; it is not derived, and a reader could propose a finer or coarser stack without being wrong. The stack is offered as a working structure that has proven useful, not as a closed result, and the lineage column throughout indicates inspiration, not derivation.

Contents

Preface: From the Coordinate to the Ground

The first three volumes (Volume 1, 2, and 3) built a coordinate and an engine that act on the structure of a claim.. The foundational and observational volumes position a claim and observe its modal stance; the analytical volume audits the inferential and control operation it performs. All of this is internal in a precise sense: it asks whether a claim is well formed and whether its operation is warranted, not whether the world it speaks of is as it says. A claim can be perfectly positioned, correctly observed, and analytically licensed, and still be socially wrong — ungrounded in the body that must act on it, the dialogue that must receive it, the organization and economy it moves through, the law it must obey, and the wider system it must not destabilize.

Axis 3 is the axis of that grounding. Where Axis 1 proposes and Axis 2 constrains, Axis 3 grounds: it checks a candidate output against the structure of the human world at six scales, from the somatic to the universal. This volume specifies that check. It is not a further subdivision of the 36/144/864 coordinate — it does not multiply the cells — but a parallel axis, orthogonal to the coordinate, that supplies the grounding the engine needs in order to act. Concretely, the grounding score the analytical engine's attenuation operator consumes, and the reference its control loop is held to, are produced here.

The stack is read from the body outward. The somatic layer asks whether an output coheres with the physical and ergonomic facts of an embodied agent. The protocol and organizational layers ask whether it coheres with communicative and organizational scope. The economic and governance layers ask whether it respects resource and value equivalence and the norms and laws of its institutions. The universal layer asks whether it keeps faith with the coherence of the wider system it belongs to. An output grounded at all six is, in the sense this volume makes precise, socially aligned.

A reader who wants the normative core may read Chapters 3 through 10 with Appendix A; the surrounding chapters supply motivation, lineage, application, and — with more candor than elsewhere in the series — the limits of what is here proposed..

Part I

Foundations of the Grounding Stack

1. Axis 3 in the HNS Full Stack

The Social Meta Layers are the third axis of the Multi-Axis Verification account. This chapter fixes their place in the full stack and states the one structural fact that governs everything else: Axis 3 is orthogonal to the coordinate of Axis 2.

1.1 The three axes

Axis 1 (generative) proposes a continuation; it is the model's native fluency. Axis 2 (structural / causal) is the 36/144/864 coordinate of the first three volumes; it positions a claim, observes its modality, and audits the operation it performs. Axis 3 (grounded / executive) is the stack specified here; it checks a candidate output against the structure of the human world. Reliability, on the MAV account, is the conjunction of the three: an output must be fluent, structurally warranted, and grounded. No axis suffices alone, and the axes are approximately independent, which is why their combination adds information that none supplies by itself.

1.2 Orthogonality: Axis 3 is not a multiplicative factor

It must be stated plainly, because the cardinalities of the earlier volumes invite the wrong inference. The Social Meta Layers do not multiply the coordinate. There is no 864 × 6 space. The layers are a parallel grounding axis: they evaluate an output that has already been positioned and analysed, and return a grounding verdict, rather than subdividing the positions. The coordinate says what a claim is and what operation it performs; the stack says whether the resulting output is grounded in the world. The two are composed at runtime, not multiplied in cardinality.

Axis 1 (propose) → Axis 2 (position, observe, analyse) → Axis 3 (ground) → emit

1.3 What Axis 3 supplies to Axis 2

The connection to the analytical engine is concrete. The engine's output-attenuation operator scales a candidate by a structural grounding score; part of that score is the social grounding computed here. And the engine's control loop holds a perceived grounding to a reference; the reference, and the higher loops that set it, are supplied by this stack. In the language of perceptual control, the Social Meta Layers are the higher controllers whose outputs become the references of the structural loop below. Axis 3 is therefore not a passive checker but the source of the targets the rest of the system controls toward.

2. Society-Aligned Artificial Intelligence

This chapter states what the stack is for: a definition of social alignment stronger than value alignment, and the six social error domains the stack is built to catch.

2.1 The grounding function

A socially aligned output is defined as one that is grounded at every layer of the stack. Writing G for the composite grounding function and g_i for the grounding verdict at layer i:

$$G(x) = 1 \text{ if and only if } g_i(x) = 1 \text{ for every layer } i = 1 \dots 6$$

An output that fails any single layer is socially misaligned on that layer, whatever its merits elsewhere. The conjunction is deliberate: groundedness in dialogue does not excuse a violation of law, and economic coherence does not excuse a physical impossibility. The stack is demanding by construction, and the next chapters specify each conjunct.

2.2 Society alignment versus value alignment

Society alignment differs from the more familiar value alignment. Value alignment asks whether an output matches individual human preferences or principles. Society alignment asks whether it respects the structural properties of the social systems in which it acts — not merely what individuals prefer, but what their institutions, norms, and shared infrastructures require. It is the stronger requirement, and it is the one that current governance frameworks implicitly demand — lawful, non-destabilizing, institutionally appropriate output — without specifying it technically. The stack is an attempt at that specification.

2.3 The six social error domains

Each layer corresponds to a way an output can be socially wrong while being individually coherent. A somatic failure violates the physical and ergonomic facts of an embodied agent. A protocol failure violates communicative alignment. An organizational failure violates the scope of the group or institution it addresses. An economic failure violates resource or value equivalence. A governance failure violates a norm or a law. A universal failure violates the coherence of the wider system. These are the targets; the layers are the checks.

3. The Six Social Meta Layers

This chapter presents the stack as a whole; Part II takes each layer in turn. The table is normative and is reproduced in Appendix A.

	Layer	Grounds	Suppresses	Lineage
SMS-1	Somatic	Bodily / ergonomic coherence	Metaphor Contamination	Barsalou; Harnad
SMS-2	Protocol	Communicative alignment	Scope Drift	Habermas
SMS-3	Organizational	Organizational scope	Scope Drift	HNS Extension
SMS-4	Economic	Resource / value equivalence	Unsupported Causality	HNS Extension
SMS-5	Governance	Normative / legal conformance	Category Ambiguity	Institutional theory
SMS-6	Universal	Global systemic coherence	Layer Jump	Ashby; GST

> ***Note.*** *Naming. "Social Meta Layers" is the axis (the six layers); "Social Meta Structure (HNS-SMS-6)" is this specification. "Social Meta" is the umbrella for the whole stack and does not assert that every layer — notably the somatic base — is itself social. SMS-3 was formerly labelled "Ecosystem"; it is renamed "Organizational" because its content is organizational scope, which the earlier label obscured.*

3.1 Reading the stack from the body outward

The layers are ordered, broadly, by scale: from the individual body (SMS-1), through dyadic and organizational interaction (SMS-2, SMS-3), to markets and institutions (SMS-4, SMS-5), to the civilizational whole (SMS-6). The ordering is a reading aid, not a strict hierarchy; as the opening note warned, scale and domain do not perfectly coincide, and the layers overlap. What the ordering does capture is the widening of the context an output must cohere with, from the agent that utters it to the system that must absorb it.

3.2 A legacy grouping

An earlier presentation grouped the six into three architectural sub-layers — AnthroDesign (human-centred design), Synaptics (alignment and correction), and NormaSphere (norms and institutions). That grouping is retained here only as a presentational convenience; the normative content is the flat set of six layers, and conformance is defined against the six, not the three.

3.3 The somatic scope note

Because the umbrella term is “social,” the somatic layer deserves a word. SMS-1 is not a social layer; it is the embodied, physical base on which the social layers rest. It is included in the stack because grounding begins in the body — an output that is physically or ergonomically impossible is ungrounded before any social question arises — not because the body is itself a social institution. The stack is “social meta” as a whole; its base is somatic.

Part II

The Grounding Layers in Detail

3.4 One output through all six layers

To see the stack work, read one output through all six. “Tell every team to cut their cloud spend by 30% this quarter to hit the savings target.” SMS-1: physically possible — passes. SMS-2: clear and on-topic — passes. SMS-3: does the speaker have authority to direct every team? — flags unless mandated. SMS-4: is a 30% cut equivalent to the target, and what capability does it cost? — flags until the balance is shown. SMS-5: does any contract or policy fix the spend? — conditional. SMS-6: if every team cuts identically, could a shared service fall below viability? — flags the systemic risk. The composite is ungrounded until the authority, the cost equivalence, and the systemic effect are addressed — a single output, six independent checks, one conjunctive verdict.

4. SMS-1 — Somatic

The first layer is the body. It grounds an output in the physical and ergonomic facts of an embodied agent acting in a physical environment, and it suppresses the failure of Metaphor Contamination — the drift in which a figure of speech is treated as a literal mechanism.

4.1 What it grounds

Somatic grounding asks whether an output is compatible with the body that must enact or receive it: with reach, force, timing, perception, fatigue, and the physical constraints of the tools and spaces involved. An instruction that ignores these — that asks for an impossible posture, an unattainable rate, a perception the senses cannot deliver — is ungrounded at SMS-1 however fluent and well intentioned it is.

4.2 Lineage

The layer draws on embodied or grounded cognition (Barsalou) and on the symbol-grounding problem (Harnad): the recognition that symbols acquire meaning by connection to sensorimotor experience, not by relation to other symbols alone. An ungrounded symbol, manipulated only against other symbols, is exactly what produces metaphor mistaken for mechanism. SMS-1 is the requirement that the chain of meaning touch the body somewhere.

4.3 The failure it suppresses

Metaphor Contamination is the treatment of a metaphor as if it were a literal causal claim — “momentum” in markets read as Newtonian inertia, “memory” in a system read as recollection. By requiring a somatic anchor, SMS-1 flags claims whose only support is a bodily metaphor doing illicit literal work, and asks that the literal mechanism be stated or the figure be marked as a figure.

4.4 A worked grounding (non-normative)

An assistant tells a warehouse worker to “keep lifting at the same pace through the double shift.” Positioned and analysed, the instruction may be perfectly well formed. At SMS-1 it fails: sustained lifting at a fixed pace ignores fatigue and injury limits of the body that must perform it. The grounding verdict $g_1 = 0$, and the output is attenuated or revised toward a physically sustainable instruction.

4.5 Boundary with the layers above

The somatic layer grounds in the body, not in any social relation, and that is what distinguishes it from every layer above. An output can be somatically grounded and socially ungrounded — physically possible yet communicatively or legally wrong — and the reverse. The layer's characteristic confusion is with metaphor: when a bodily figure such as “grasp,” “weight,” or “pressure” is read literally, the check at SMS-1 asks whether a real body is involved or only its image.

5. SMS-2 — Protocol

The second layer is communication. It grounds an output in the conditions of successful communicative exchange and suppresses Scope Drift — the silent broadening or shifting of what a claim is about. It anchors, in the Natural Layers of Axis 2, the micro-social layer L5.

5.1 What it grounds

Protocol grounding asks whether an output is intelligible and appropriate as a communicative act between agents: whether it can be understood by its addressee, whether it raises claims it is prepared to redeem, and whether it stays within the topic and commitments of the exchange. An output that quietly changes the question, over-claims what was asked, or addresses the wrong interlocutor fails here.

5.2 Lineage

The layer draws on Habermas's theory of communicative action, in which an utterance oriented to understanding implicitly raises validity claims — to truth, to normative rightness, and to sincerity — that the speaker must be ready to redeem in dialogue. SMS-2 is the requirement that an output be defensible as such an act: that its implicit claims be ones the system could, if challenged, make good.

5.3 The failure it suppresses

Scope Drift is the unannounced movement of a claim's scope — answering a narrower or wider question than was asked, or letting the subject slide across a conversation. SMS-2 suppresses it by holding the output to the communicative frame of the exchange. The suppression is shared with SMS-3: drift across an organizational boundary is caught at the organizational layer, drift within a dialogue at the protocol layer.

5.4 A worked grounding (non-normative)

Asked "is this contract clause enforceable in our jurisdiction?" a system answers with a general essay on contract law. Communicatively, it has drifted: it answers a broader question than the one posed and leaves the addressee's actual question unaddressed.

$g_2 = 0$; the output is redirected to the specific clause and jurisdiction asked about.

5.5 Boundary with the organizational layer

Protocol and organizational grounding are easy to conflate, because a dialogue is often also an organizational act. The boundary is the kind of frame violated: SMS-2 concerns the communicative frame between the parties to an exchange — intelligibility, topic, redeemable claims — while SMS-3 concerns the organizational frame of scope, role, and authority. The same utterance can pass one and fail the other: clear and well formed (SMS-2) yet beyond the speaker's mandate (SMS-3). The shared suppression of Scope Drift is exactly this overlap made visible.

6. SMS-3 — Organizational

The third layer is the organization. It grounds an output in the scope and structure of the group or institution it addresses, and it shares with SMS-2 the suppression of Scope Drift. Like the protocol layer, it anchors the micro-social layer L5 of Axis 2.

6.1 What it grounds

Organizational grounding asks whether an output respects the scope, roles, and authority structure of the organization in which it acts: whether it addresses the right level, stays within the mandate of the unit it speaks for, and does not assume coordination or authority that the organization does not grant. An output that issues an instruction beyond its remit, or assumes a coordination the organization cannot deliver, fails here.

6.2 Lineage — an HNS extension

This layer is, candidly, an HNS extension with no single external proponent. It draws loosely on organizational theory and on the distinction between individual interaction and collective structure, but it is not the faithful application of one named framework. It is included because organizational scope is a real and distinct ground of failure between the dyadic (SMS-2) and the institutional (SMS-5), and the stack would have a gap without it. Its status as a designed rather than inherited layer is marked here and in the lineage table.

6.3 The failure it suppresses

Organizational Scope Drift is the broadening of a claim across an organizational boundary — treating a team's decision as the company's, a department's data as the whole organization's, an advisory remark as an authoritative instruction. SMS-3 suppresses it by holding the output to the scope the organization actually grants. That Scope Drift is suppressed jointly by SMS-2 and SMS-3, rather than by one layer alone, is the clearest case in the stack of the failure-to-layer mapping being many-to-one.

6.4 A worked grounding (non-normative)

A drafting assistant turns one engineer's suggestion into "the company has decided to adopt" a standard. Organizationally, it has overstepped: a suggestion at one scope is asserted at a higher one the organization never authorized. $g_3 = 0$; the output is rescoped to the suggestion and the level that actually made it.

6.5 Boundary with the governance layer

Organizational and governance grounding both concern collective structure, but at different bindingness. SMS-3 concerns the scope and authority an organization grants in fact; SMS-5 concerns the norms and laws that bind it from outside. An action can be within organizational scope, so SMS-3 passes, yet unlawful, so SMS-5 fails; or lawful, so SMS-5 passes, yet beyond the unit's mandate, so SMS-3 fails. The two are checked separately for this reason.

7. SMS-4 — Economic

The fourth layer is the economy. It grounds an output in resource and value equivalence — the requirement that what is claimed to be exchanged, produced, or saved actually balances — and it suppresses Unsupported Causality at the level of economic mechanism.

7.1 What it grounds

Economic grounding asks whether an output respects conservation and equivalence: whether resources claimed to be created were in fact transferred from somewhere, whether costs and benefits balance, whether a proposed gain has a stated source. An output that promises value from nowhere — savings without a cost moved, output without an input — fails here, in the way a physical claim that violated conservation would fail at SMS-1.

7.2 Lineage — an HNS extension

Like the organizational layer, the economic layer is an HNS extension rather than the application of a single proponent's framework. It draws on the general intuition of equivalence and conservation that runs through economic thought, but it is a designed layer, included because resource and value coherence is a distinct ground of failure not covered by the layers around it. Its designed status is marked.

7.3 The failure it suppresses

At this layer, Unsupported Causality takes an economic form: the assertion that an action produces a value effect without an accounted mechanism — "this change will increase revenue" with no stated source of the increase and no cost it displaces. SMS-4 suppresses it by demanding the equivalence: where does the value come from, and what does it cost? The check complements the analytical engine's general treatment of Unsupported Causality with the specific demand for an economic balance.

7.4 A worked grounding (non-normative)

A plan claims that a new process will "save time for everyone with no added work." Economically, the equivalence is missing: the saved time must be paid for somewhere — setup, maintenance, a shifted burden. $g_4 = 0$ until the source and cost are stated; the output is revised to name where the saving comes from.

7.5 Boundary with causal analysis

Economic grounding is distinguished from the causal analysis of Axis 2 by its specific demand. The analytical engine asks whether a causal claim is warranted in general; SMS-4 asks the narrower, checkable question of equivalence — does the value balance? A claim can pass the general causal check and still fail SMS-4 if it conjures value without a source. The economic layer is, in effect, conservation applied to value, and that is what makes it mechanically checkable rather than a matter of judgement.

8. SMS-5 — Governance

The fifth layer is governance. It grounds an output in normative and legal conformance and suppresses Category Ambiguity — the confusion of one normative category with another. It anchors the macro-social layer L6 of Axis 2 and connects directly to Volume 6 (Standards & Conformance Specification).

13.1 Volume 6 develops this into a regulatory mapping.

8.1 What it grounds

Governance grounding asks whether an output conforms to the norms, rules, and laws of the institutions it acts within: whether it respects rights and obligations, stays within what regulation permits, and keeps normative categories distinct — advice from instruction, permission from requirement, recommendation from ruling. An output that issues as a requirement what is only a recommendation, or treats the permitted as the mandatory, fails here.

8.2 Lineage

The layer draws on institutional theory — the study of how norms, rules, and conventions constitute and constrain action. What institutional theory supplies is the recognition that conformance is structural, not merely preferential: institutions define categories of permitted, required, and forbidden, and a competent actor keeps these straight. SMS-5 is the requirement that an output keep them straight.

8.3 The failure it suppresses

Category Ambiguity, at the governance layer, is the blurring of normative categories — reading “may” as “must,” “guidance” as “law,” a private norm as a public rule. SMS-5 suppresses it by holding the output to the correct normative category, in the same spirit that the observational layer of Axis 2 holds a claim to a single modal quadrant. The two reinforce each other: a modal error at Q-level often surfaces as a governance category error here.

8.4 A worked grounding (non-normative)

A compliance assistant states that staff “must” take an optional training. The normative category is wrong: a recommendation is asserted as a requirement. $g_5 = 0$; the output is corrected to mark the training as recommended, not mandatory, unless a rule in fact requires it.

8.5 Boundary with the observational layer

Governance grounding and the observational layer of Axis 2 both police category. The observational layer keeps a claim to one modal quadrant — subjective or objective, necessary or contingent; SMS-5 keeps it to one normative category — permitted, required, or forbidden. The two often co-occur: a claim that slides from “may” to “must” is both a modal drift on Axis 2 and a governance category error on Axis 3. SMS-5 is the social face of the same discipline.

9. SMS-6 — Universal

The sixth layer is the system as a whole. It grounds an output in global systemic coherence — the requirement that an output not destabilize the wider system it belongs to — and it suppresses Layer Jump, the illegitimate leap across levels of organization.

9.1 What it grounds

Universal grounding asks whether an output is compatible with the optimality and stability of the whole: whether a locally sensible action would, if generalized, undermine the system that makes it sensible; whether it respects the constraints that keep the larger structure viable. An output that is coherent at every lower layer but corrosive in aggregate — a recommendation harmless once and ruinous universally — fails here.

9.2 Lineage

The layer draws on general systems theory and on Ashby's cybernetics — in particular the idea that a viable system maintains its essential variables within limits, and that a regulator must have variety enough to meet the disturbances it faces. This lineage was chosen over a scale-named alternative ("civilization studies") deliberately: the lineage column should anchor the layer's function — systemic viability — rather than its scale. The universal layer is about keeping a system coherent, and systems theory is the study of exactly that.

9.3 The failure it suppresses

Layer Jump is the illegitimate transfer of a claim across levels of organization — reasoning from an individual case to a systemic law, or from a systemic regularity to an individual prescription, without the bridge each requires. SMS-6 suppresses it by checking an output against the coherence of the level it actually addresses and the level it would affect. It is the macro complement to the layer-adjacency discipline of the foundational volume.

9.4 A worked grounding (non-normative)

A system advises every client to exploit the same arbitrage. Locally rational, the advice is universally incoherent: if all act on it the arbitrage closes and the wider market is destabilized. $g_6 = 0$; the output is qualified to the conditions under which it remains systemically harmless.

9.5 Boundary with the layers below

Universal grounding is distinguished from every layer below by a quantifier. The lower layers ask whether this output is grounded; SMS-6 asks whether it would remain grounded if generalized across the system. An output can pass every lower layer for the individual case and fail SMS-6 in aggregate — the locally rational, globally ruinous recommendation. It is the only layer whose check ranges over the population of similar acts rather than the single act in hand.

Part III

The Grounding Function in Operation

10. The Grounding Function G(x)

This chapter specifies how the six per-layer verdicts compose into a single grounding result and how that result enters the runtime of the analytical engine.

10.1 Composition

Each layer returns a grounding verdict for a candidate output. In the strict form the verdict is binary, $g_i \in \{0, 1\}$, and the composite is their conjunction — the output is grounded only if grounded everywhere:

$$G(x) = g_1(x) \wedge g_2(x) \wedge \ldots \wedge g_6(x)$$

In the graded form used by the runtime, each layer returns a margin $g_i \in [0, 1]$, and the composite is a conservative aggregate — the minimum, so that a single weak layer pulls the whole grounding down, or a weighted product when a softer combination is wanted:

$$s = \min_i g_i(x) \quad \text{or} \quad s = \prod_i g_i(x)^{w_i}, \ \Sigma\, w_i = 1$$

Both aggregates are monotone non-decreasing in each layer's margin: improving grounding on any layer never lowers the composite. This is the property the analytical engine's attenuation operator relies on, and it mirrors that engine's own composition of operator-support margins.

10.2 Feeding the analytical engine

The composite grounding s produced here is the social component of the structural grounding score the analytical engine's output-attenuation operator consumes. Where that engine scales a candidate by g(s), the s it scales by carries the verdict of this stack. The connection is exact: a low social grounding lowers s, and a low s attenuates the output, so a socially misaligned continuation is suppressed by the same mechanism that suppresses an analytically unsupported one. The stack diagnoses; the engine's actuator acts.

10.3 Setting references

The stack also supplies the references the engine's control loop is held to. In a control hierarchy the output of a higher loop sets the reference of a lower one; here the Social Meta Layers are the higher loops, and the grounding target they define becomes the reference the structural loop controls toward. Governance and universal grounding, in particular, set references that the lower layers must not violate — a lawful, systemically coherent target that the rest of the system steers to meet.

10.4 A worked composite (non-normative)

Suppose an output returns per-layer margins $g_1 = 1.0$, $g_2 = 1.0$, $g_3 = 0.4$, $g_4 = 0.5$, $g_5 = 1.0$, $g_6 = 0.8$ — grounded somatically and communicatively, weak organizationally and economically. Under the conservative aggregate s = min = 0.4 the weakest layer governs: the output is carried at four-tenths strength, so its organizationally over-scoped, economically unbalanced phrasing falls below grounded alternatives in the engine's pre-decode reweighting. Under a uniform-weight product, s = (1·1·0.4·0.5·1·0.8)^{1/6} ≈ 0.74, the same output is treated more leniently. The choice between the two is the stack's main tuning decision: the minimum refuses any single ungrounded layer, while the product tolerates one weak layer if the rest are strong. Both are monotone, so improving any layer never lowers s.

11. The Theoretical Basis of Grounding

The stack is not an arbitrary list of six concerns; it is built on a coherent picture of how an adaptive system stays coupled to its environment. This chapter states that picture and names its sources, with the usual caution that they are inspiration, not derivation.

11.1 Prediction, control, and free energy

Two convergent traditions underlie the stack. Predictive coding and the free-energy principle (Rao and Ballard; Friston) describe an agent that maintains itself by minimizing the discrepancy between its predictions and its sensed world, acting and updating so as to stay within viable bounds. Perceptual control theory (Powers) describes the same coupling as the control of perception toward references. The Social Meta Layers extend this coupling outward: the references an agent must meet are not only sensory but social, and grounding is the minimization of discrepancy against the structure of the human world at six scales.

11.2 Cybernetics and viability

The outermost layer rests on cybernetics and general systems theory (Wiener; Ashby). A viable system holds its essential variables within limits, and a regulator must command variety enough to counter the disturbances it meets. Universal grounding is this requirement applied to an AI's outputs: that they not drive the system they act in outside its viable range. The stack thus runs from the body, where grounding is sensorimotor, to the civilization, where grounding is systemic viability — one principle, coupling to environment, read at widening scales.

11.3 From body to civilization

The arc of the stack is deliberate. SMS-1 grounds in the body; SMS-2 and SMS-3 in interaction and organization; SMS-4 and SMS-5 in markets and institutions; SMS-6 in the civilizational whole. The claim is not that these scales are discrete or that exactly six exist — the candor of the opening note stands — but that grounding must be checked across this range, because an output can be coupled to the body and uncoupled from the institution, or coherent in the institution and corrosive to the system. The stack is the attempt to check the whole range rather than one scale of it.

12. Failure Suppression and the Social Error Domains

This chapter states the mapping between the layers and the failures they suppress, and is explicit that the mapping is not one-to-one.

Layer	Social error domain	Suppressed failure
SMS-1 Somatic	Physical / ergonomic violation	Metaphor Contamination
SMS-2 Protocol	Communicative misalignment	Scope Drift (with SMS-3)
SMS-3 Organizational	Organizational over-scope	Scope Drift (with SMS-2)
SMS-4 Economic	Resource / value imbalance	Unsupported Causality
SMS-5 Governance	Normative / legal nonconformance	Category Ambiguity
SMS-6 Universal	Systemic incoherence	Layer Jump

12.1 The mapping is many-to-one

The clearest departure from a tidy correspondence is Scope Drift, suppressed jointly by the protocol and organizational layers. Drift within a dialogue is a protocol failure; drift across an organizational boundary is an organizational one; the same named failure is caught at two layers depending on where the boundary it crosses lies. The lesson generalizes: the layers are checks on grounding, and a single failure type can be ungrounded in more than one way. The mapping is offered as a guide to where to look, not as a partition.

12.2 Detection and response

At runtime each layer returns its margin; a margin below the layer's reference is a grounding shortfall in that layer's error domain. The shortfall lowers the composite grounding s, and the analytical engine's attenuation acts on the lowered s. As elsewhere in the framework, detection is structural and the response is regulative, and both are external to the generating model and written to the audit record.

12.3 A combined detection (non-normative)

A single output can fail at more than one layer, and the stack records each. “Because customers love the new design, we must roll it out company-wide immediately, and it will pay for itself.” The claim fails at SMS-2 and SMS-3 (Scope Drift: a preference of some customers asserted as a company-wide mandate), at SMS-4 (Unsupported Causality: “pay for itself” with no stated source), and at SMS-5 (Category Ambiguity: “must” where only “may” is supported). The composite is low on three layers at once, and the record shows each, so the correction can be directed at each in turn — rescope the claim, state the economic balance, and fix the normative category.

Part IV

Application, Lineage, and Status

13. Applications

The stack is meant to be used. This chapter sketches three settings, briefly, with the caveat that each is an application of a proposed structure, not a validated result.

13.1 AI governance

For governance, the stack offers what process-based frameworks lack: a technical place to attach a conformance check. An output can be evaluated against the governance and universal layers for legal conformance and systemic coherence, and the per-layer verdict logged, so that "compliant" becomes a recorded, auditable property of each output rather than a property asserted of the system as a whole. The sixth volume develops this into a regulatory mapping.

13.2 Organizational and institutional use

Within an organization, the protocol and organizational layers check that an assistant's outputs stay within communicative and organizational scope — that they address the right level, claim the right authority, and respect the mandate of the unit they speak for. This is the everyday face of the stack: most socially misaligned outputs in practice fail at SMS-2 or SMS-3, not at the dramatic extremes.

13.3 Civilization-scale considerations

At the largest scale, the universal layer asks the question that individual alignment cannot: not whether an output is good for its user, but whether it would remain coherent if generalized across the system. This is the layer that catches the locally rational, globally ruinous recommendation, and it is the layer most in need of the empirical work the final chapter calls for.

14. Intellectual Lineage and Proponents

This chapter records the debts and, with the candor the stack especially requires, distinguishes the inherited layers from the designed ones.

14.1 The inherited layers

SMS-1 draws on embodied cognition (Barsalou) and the symbol-grounding problem (Harnad). SMS-2 draws on Habermas's communicative action. SMS-5 draws on institutional theory. SMS-6 draws on general systems theory and Ashby's cybernetics. The theoretical basis of the whole — prediction, control, and viability — draws on Friston, Rao and Ballard, Powers, Wiener, and Ashby. Each is acknowledged as inspiration for a layer's function, not as a framework the layer faithfully implements.

14.2 The designed layers

SMS-3 (Organizational) and SMS-4 (Economic) are HNS extensions with no single external proponent. They are included because organizational scope and economic equivalence are real and distinct grounds of failure that the inherited layers leave uncovered, but their inclusion is a design judgement, and they are the layers most open to revision. Marking them as designed rather than inherited is part of the honesty the stack is held to.

Layer	Source	Inherited or designed
SMS-1 Somatic	Barsalou; Harnad	inherited
SMS-2 Protocol	Habermas	inherited
SMS-3 Organizational	—	designed (HNS)
SMS-4 Economic	—	designed (HNS)
SMS-5 Governance	Institutional theory	inherited
SMS-6 Universal	Ashby; general systems theory	inherited

14.3 What is HNS's own

The assembly of these six concerns into a single grounding axis, the composite grounding function, the two designed layers, and the use of the stack as the source of the grounding score and references that drive the analytical engine are HNS's own. The proponents named did not design the stack and should not be read as endorsing it; the recombination, and the responsibility for it, are the author's.

15. Empirical Status and Limitations

This volume closes with the most candid limitations chapter in the series, because the stack is the least settled part of the framework.

15.1 What is established, proposed, and conjectured

Established are the source traditions and the well-definedness of the grounding function: the conjunction and the monotone aggregates are elementary, and every per-layer verdict is externally auditable. Proposed, and defensible but unproven, are the six layers themselves, the two designed layers especially, and the claim that this is a good decomposition of social grounding. Conjecture are the practical claims — that checking against the stack measurably improves social alignment in deployment.

15.2 The stack is a scaffold, not a proven taxonomy

The candor promised at the outset is owed in full here. The stack mixes scale and domain: it reads as a progression from body to civilization, yet its middle layers are distinguished by domain — communicative, organizational, economic, normative — and these do not line up neatly along the scale. The layers are not mutually exclusive: the economic and governance layers overlap wherever law prices behaviour, and the protocol and organizational layers overlap wherever a dialogue is also an organizational act; the shared suppression of Scope Drift is the visible symptom. And there are gaps: an ecological layer (grounding in the non-human environment), an ethical layer distinct from the legal, and a temporal layer (grounding across time and future generations) all have a claim to inclusion that the present six do not honour. The number six is a chosen resolution that keeps this axis aligned with the rest of the framework, not a count derived from the structure of social reality.

15.3 The honest position

None of this voids the stack's usefulness. A scaffold that catches somatic impossibility, communicative and organizational over-scope, economic imbalance, legal nonconformance, and systemic incoherence catches most of the ways a fluent output is socially wrong, and it does so in an external, auditable form. But it should be used as a working structure that has proven useful, and revised as evidence accumulates — a stable but explicitly revisable v1.0, more so than any other volume in the series. The threats most worth naming are the overlap and gap problems above, the difficulty of computing per-layer grounding reliably and within a latency budget, the inter-rater reliability of the grounding verdicts, and the boundary between a structural defect (Axis 2) and a grounding defect (Axis 3), which is not always sharp.

15.4 The known gaps

Three omissions are named so that they are not mistaken for oversights. An ecological layer — grounding in the non-human environment, the resource and sink limits of the natural world — is absent; the economic layer touches it only where value is priced, leaving un-priced environmental cost ungrounded. An ethical layer distinct from the legal is absent; governance grounds in norms and laws as they are, not in whether they are right, so an output can be lawful and unethical and still pass SMS-5. And a temporal layer — grounding in future states and generations — is absent; the universal layer reaches it only through present systemic viability, not through an explicit obligation to the future. Each is a candidate further layer, and each is a reason the present six should be read as a working scaffold rather than a closed set.

11.4 Why these traditions cohere

It is fair to ask why six concerns drawn from such different fields belong on one axis. The answer is that they are one principle read at widening scales: an adaptive system stays viable by keeping its coupling to its environment within bounds. At the somatic scale that coupling is sensorimotor; at the communicative and organizational scales it is the alignment of an act with the parties and structures it engages; at the economic and governance scales it is conformance to equivalence and to institutional rule; at the universal scale it is the viability of the whole. Prediction, control, and requisite variety are the same idea applied at each scale. The stack is the claim that grounding must be checked across the whole range, because coupling can hold at one scale and fail at another.

Appendix A — Normative Reference Tables

The tables below are normative. In case of conflict between prose and table, the table governs; in case of conflict with the Unified Structural Coordinate Table, that table governs.

A.1 The six Social Meta Layers

	Layer	Grounds	Suppresses	Lineage
SMS-1	Somatic	Bodily / ergonomic coherence	Metaphor Contamination	Barsalou; Harnad
SMS-2	Protocol	Communicative alignment	Scope Drift	Habermas
SMS-3	Organizational	Organizational scope	Scope Drift	HNS Extension
SMS-4	Economic	Resource / value equivalence	Unsupported Causality	HNS Extension
SMS-5	Governance	Normative / legal conformance	Category Ambiguity	Institutional theory
SMS-6	Universal	Global systemic coherence	Layer Jump	Ashby; GST

A.2 The grounding function

Strict: $G(x) = 1$ iff $g_i(x) = 1$ for all i. Graded runtime: $s = \min_i g_i(x)$, or a weighted product; both monotone non-decreasing in each layer. The composite s is the social component of the structural grounding score consumed by the analytical engine's output-attenuation operator (Volume 3, O6).

A.3 Axis placement

Axis	Role	Relation to the coordinate
Axis 1	Generative / statistical	proposes
Axis 2	Structural / causal (36/144/864)	the nested coordinate
Axis 3	Grounded / executive (SMS-6)	parallel, orthogonal — not a multiplicative factor

Appendix B — Per-Layer Grounding Checklist

A non-normative template for evaluating a candidate output against the stack. Each layer is checked independently; the composite follows Appendix A.2.

- SMS-1 Somatic: is the output physically and ergonomically possible for the body that must enact or receive it?
- SMS-2 Protocol: is it intelligible and appropriate as a communicative act, and does it stay within the topic and commitments of the exchange?
- SMS-3 Organizational: does it respect the scope, roles, and authority of the organization it addresses?
- SMS-4 Economic: does what it claims to exchange, produce, or save actually balance, with a stated source and cost?
- SMS-5 Governance: does it conform to the relevant norms and laws, and keep normative categories (advice/instruction, may/must) distinct?
- SMS-6 Universal: would it remain coherent if generalized across the system it acts in?

Appendix C — Worked Groundings

Non-normative end-to-end examples, each an output evaluated against the six layers.

“To cut costs, tell all suppliers we will pay in 90 days instead of 30.” SMS-1 passes (physically possible). SMS-2 passes (clear). SMS-3 depends on authority to set payment terms. SMS-4 flags: the “saving” is a cost shifted to suppliers, not value created — the equivalence must be stated. SMS-5 flags if contracts or law fix the terms. SMS-6 flags if, generalized, the practice would destabilize the supplier base. Composite: grounded only once the shifted cost, the contractual position, and the systemic effect are addressed.

“Reply to the customer that the refund is guaranteed.” SMS-2 passes. SMS-5 is decisive: “guaranteed” asserts a normative category (a binding commitment) that may exceed policy — a Category Ambiguity. If policy makes the refund discretionary, $g_5 = 0$ and the output is corrected to the category policy supports.

Appendix D — Correspondence with Axis 2 and EVA

This appendix records how the stack connects to the rest of the framework.

D.1 Anchoring to the Natural Layers

The social layers anchor the upper Natural Layers of Axis 2: SMS-2 and SMS-3 anchor the micro-social layer L5; SMS-5 and SMS-6 anchor the macro-social layer L6. The somatic layer SMS-1 anchors the physical and cellular base, L1–L2. The correspondence is one of grounding, not identity: the Natural Layer is where a claim originates within the coordinate, and the Social Meta Layer is the external structure that claim must be grounded against.

D.2 Feeding the analytical engine and EVA

The composite grounding s produced by the stack is the social component of the score the analytical engine's attenuation operator consumes, and the references the stack sets become the targets that engine's control loop holds to. The per-layer verdicts are written to the external verification record specified in Volume 5 (External Verification Architecture), so that an output's social grounding can be audited by a third party who never sees the model. The stack is the grounding source; the external verification architecture is the recorder and enforcer.

Appendix E — Social Error Casebook

Ten non-normative cases, chosen to cover the six layers and the way a single output can fail several at once.

E.1 Somatic impossibility (SMS-1). "Inspect all 4,000 units by hand before noon." Physically impossible for the workforce and time given. Verdict: $g_1 = 0$; revise to a feasible rate or sample.

E.2 Metaphor as mechanism (SMS-1). "The team has lost momentum, so apply more force to speed them up." A physical metaphor is treated as a literal lever. Verdict: g_1 flags Metaphor Contamination; state the actual mechanism (workload, blockers).

E.3 Communicative drift (SMS-2). Asked whether one clause is enforceable, the system delivers a general treatise. Verdict: $g_2 = 0$ (Scope Drift within the exchange); answer the clause asked about.

E.4 Organizational over-scope (SMS-3). One engineer's suggestion is reported as "the company has decided." Verdict: $g_3 = 0$ (Scope Drift across an organizational boundary); rescope to the suggestion and its level.

E.5 Joint protocol/organizational (SMS-2+3). In a customer chat, the agent commits the company to a bespoke contract term. Clear (SMS-2 passes) but beyond mandate (SMS-3 fails). Verdict: rescope to what the agent may offer.

E.6 Value from nowhere (SMS-4). "This change saves time for everyone with no added work." No source for the saving. Verdict: $g_4 = 0$ (Unsupported Causality, economic form); name the cost or shifted burden.

E.7 Governance category error (SMS-5). Optional training is described as something staff "must" complete. Verdict: $g_5 = 0$ (Category Ambiguity, may/must); mark it recommended unless a rule requires it.

E.8 Authority versus law (SMS-3 vs SMS-5). A manager's lawful-sounding directive in fact exceeds the unit's mandate though it breaks no law. SMS-5 passes; SMS-3 fails. Verdict: the layers disagree, and both verdicts are logged.

E.9 Systemic generalization (SMS-6). Every client is advised to exploit the same arbitrage. Locally rational, universally self-defeating. Verdict: $g_6 = 0$ (Layer Jump / systemic incoherence); qualify to conditions of systemic harmlessness.

E.10 Multi-layer failure. "Customers love it, so we must ship company-wide now, and it pays for itself." Fails SMS-2/3, SMS-4, and SMS-5 at once. Verdict: low composite; correct each — rescope, balance, recategorize.

Appendix F — Correspondence with the Legacy SMS-A/B/C Grouping

An earlier architectural presentation grouped the six layers into three sub-layers. That grouping is retained only as a presentational device; the normative content is the flat set of six. The correspondence is recorded here for readers of the earlier specification.

Legacy sub-layer	Description	Canonical layers
SMS-A — AnthroDesign	Human-centred social design	SMS-1 Somatic; SMS-2 Protocol
SMS-B — Synaptics	Alignment and correction between AI and society	SMS-3 Organizational; SMS-4 Economic
SMS-C — NormaSphere	Norms, institutions, and civilization	SMS-5 Governance; SMS-6 Universal

Conformance is defined against the six canonical layers and their grounding function (Appendix A), not against the three groups. Where the legacy specification referred to a separate "SOHU" social operating-system component, that role is subsumed here by Axis 3 itself: the Social Meta Layers are the social operating layer of HNS.

Glossary

Axis 3 (Grounded / Executive) — The grounding axis: the six Social Meta Layers, checking an output against the structure of the human world.

Social Meta Layers — The six grounding layers (SMS-1 … SMS-6) that constitute Axis 3.

Social Meta Structure (HNS-SMS-6) — This specification; the document defining the stack.

Grounding function G(x) — The composite function returning 1 only if the output is grounded at every layer.

Grounding margin g_i — A layer's grounding verdict for an output, binary or in [0,1].

Society-aligned AI — An AI whose outputs satisfy G(x) = 1 — grounded against social structure, not merely individual preference.

SMS-1 Somatic — Bodily / ergonomic grounding; suppresses Metaphor Contamination.

SMS-2 Protocol — Communicative grounding; suppresses Scope Drift; anchors L5.

SMS-3 Organizational — Organizational-scope grounding (HNS extension; formerly "Ecosystem"); suppresses Scope Drift; anchors L5.

SMS-4 Economic — Resource / value-equivalence grounding (HNS extension); suppresses Unsupported Causality.

SMS-5 Governance — Normative / legal grounding; suppresses Category Ambiguity; anchors L6.

SMS-6 Universal — Systemic-coherence grounding (Ashby / GST); suppresses Layer Jump; anchors L6.

Orthogonality (Axis 3) — The property that the stack grounds output without subdividing the 36/144/864 coordinate; not a multiplicative factor.

Symbol grounding — The problem (Harnad) of how symbols acquire meaning by connection to sensorimotor experience, not to other symbols alone.

Requisite variety — Ashby's principle that a regulator must command variety enough to counter the disturbances it faces.

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